Implementation and Evaluation of AI-Assisted, Digitally Designed and 3D-Printed Transtibial Prosthetic Sockets in The Gambia

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Introduction

The World Health Organization (WHO) estimates that 90% of people requiring prosthetic and orthotic services do not have access to them, primarily due to limited awareness, service availability, trained personnel, supportive policies, and financing. In low- and middle-income countries, prosthetic sockets are typically made from manually modified plaster casts, a labor-intensive process in which quality depends heavily on the individual prosthetist's skills. Digitally designed, 3D-printed sockets in which AI-assisted shape prediction replaces manual rectification could make production faster and less dependent on scarce expertise. In The Gambia, prosthetic services are centralized within the National Rehabilitation Centre (NRC), where this technology could help address the existing gap between the demand for and availability of prosthetic care. The aim of the study is to evaluate the performance in terms of comfort and shape quality of a transtibial prosthetic socket that was manufactured using 3D scanning, digital design with AI-assisted shape prediction, and 3D printing.

Methods

This prospective interventional feasibility study aims to include 20 existing transtibial prosthesis users by the end of 2026. Residual limb geometry was captured using 3D scanning and used to digitally design patient-specific sockets using 3DMedX software from Radboud University Medical Center with AI-assisted automated shape prediction.[1] Diagnostic sockets were subsequently fabricated using 3D printing, followed by clinical fitting and shape optimisation. Definitive sockets were laminated with fiberglass and resin over the 3D-printed socket.[2] Questionnaires were administered at the first prosthesis fitting, where socket comfort scores (SCS) and socket quality scores (SQS) were measured by prosthetic users and clinicians, respectively.

Results

At the time of abstract submission, 11 of the planned 20 participants had been fitted with digitally designed prosthetic sockets. One discontinued the fitting process due to a previous negative experience with 3D-printed prostheses, while none of the remaining 10 patients required adjustments or replacement of the initial socket design. Median SCS, rated on a 0–10 scale by prosthesis users, was 10 (IQR, 8–10), while median SQS rating by prosthetists was 9 (IQR, 8–10).

Conclusions

High socket comfort scores among patients and high quality ratings from prosthetists support the feasibility of a digital workflow combining user-friendly, AI-assisted software with 3D scanning and printing technology. Inclusion is expected to be completed by the end of 2026. This approach may offer a scalable solution to help address the high unmet need for prosthetic services in The Gambia.

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References

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